

Unnamed_Unit

Unit 4 - Code Quest

PDF Documents

PDF Document 1: 1738157458746-Chapter 9 - Unit 4 - Code Quest.pdf

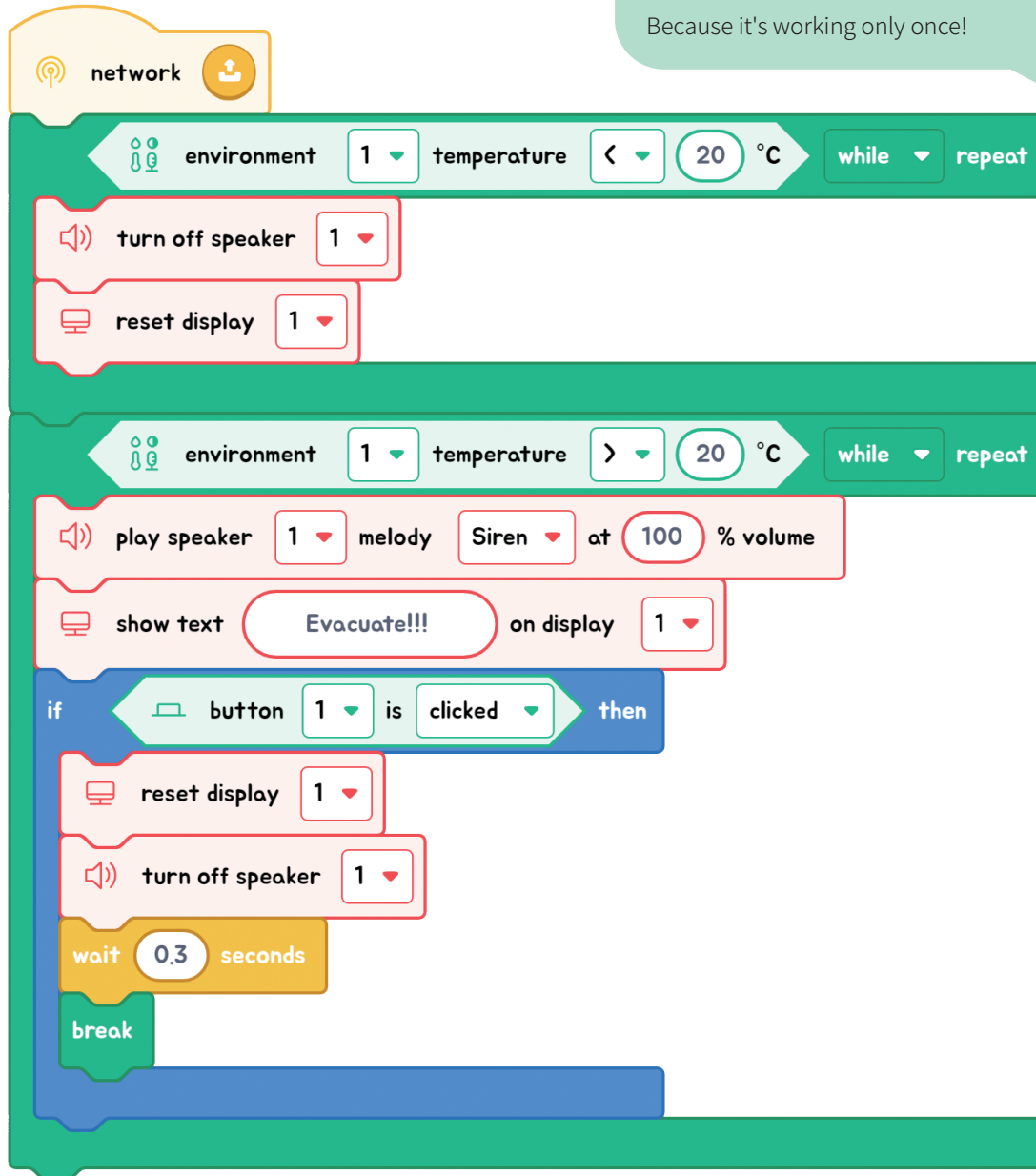
This PDF document is included in the unit materials.



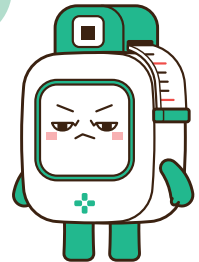
UNIT 4 CODE QUEST

Applying Knowledge (code & debugging)

CODE



This one is not covered with repeat forever!
Because it's working only once!

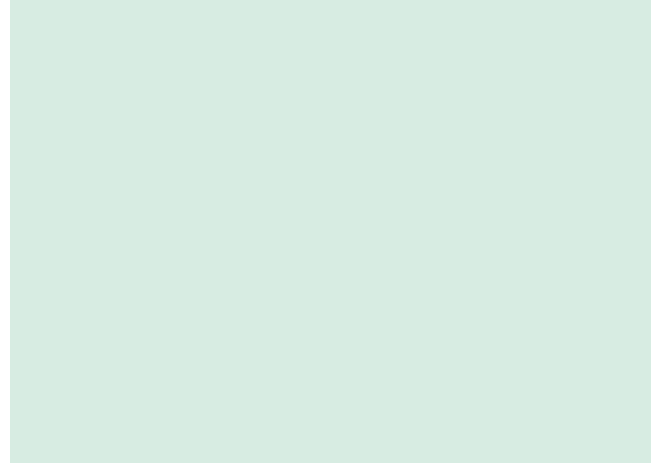
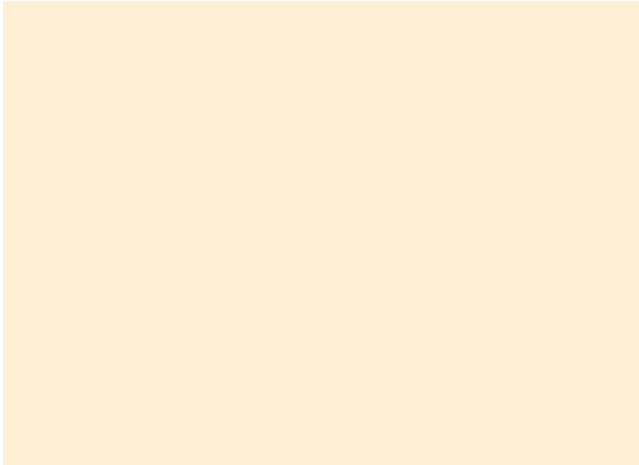


Tip!

Classroom temperatures fluctuate with the seasons. To address this, let's implement an environmental module to monitor and control the temperature. We can utilize our body heat by touching the sensor to change the temperature reading. Let's experiment to see if this method can effectively adjust the classroom temperature.

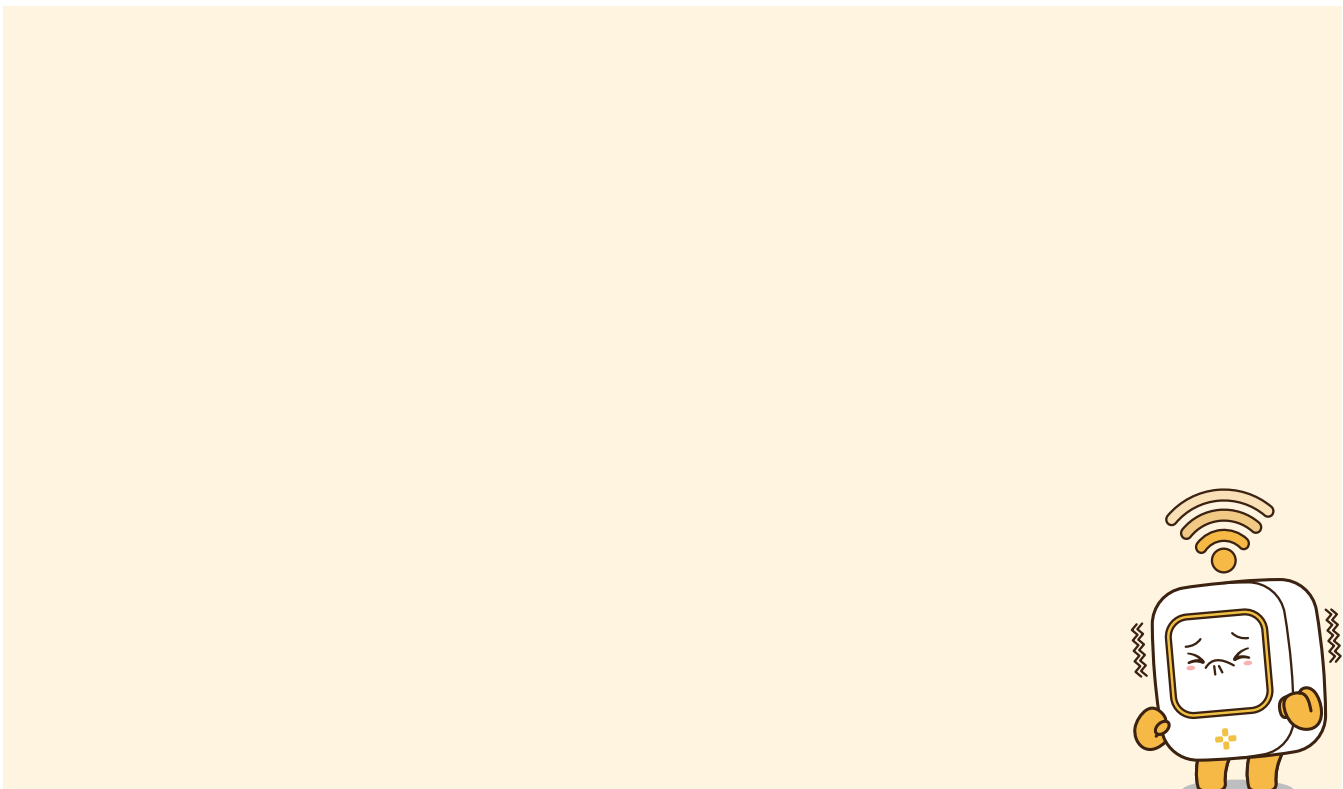
DEBUGGING

1. What will happen if we put repeat forever to cover the code?
2. What the role of break block?



MISSION 1

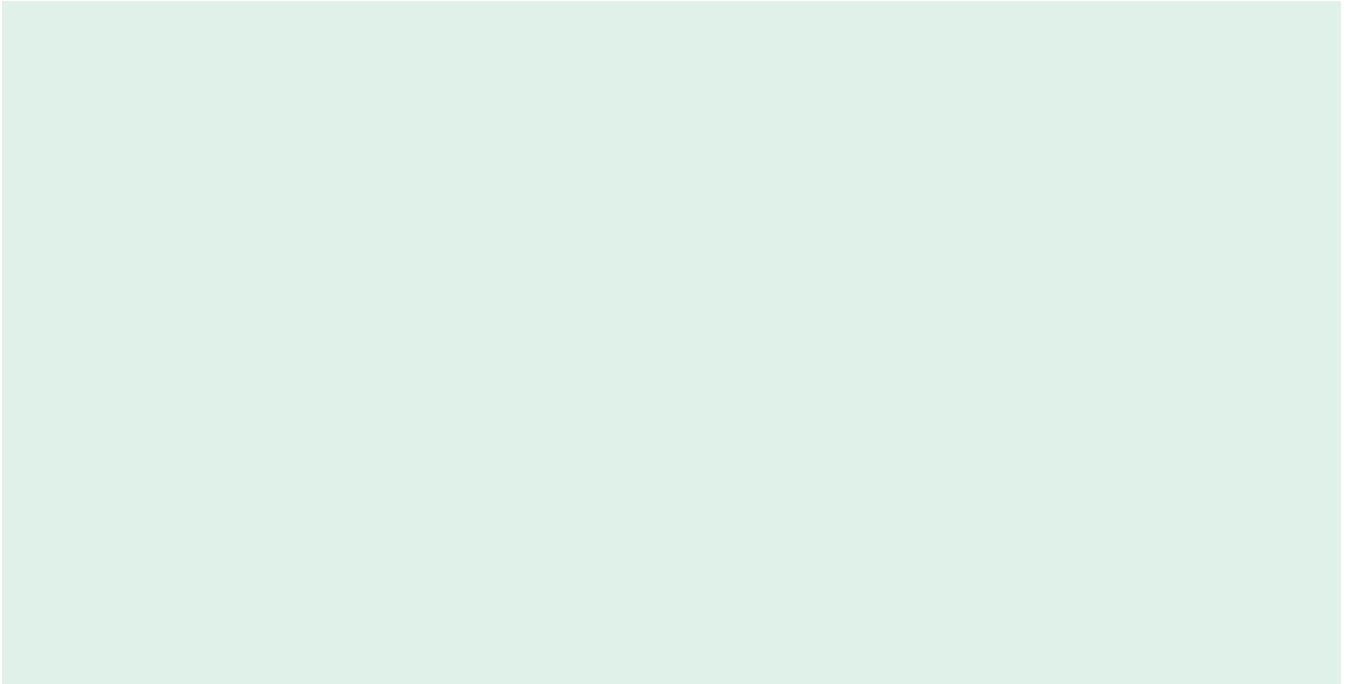
1. Since it is difficult to change the actual temperature, let's write the code using a dial instead of an Environment module. Trigger an alarm when the Dial value is above 50%.



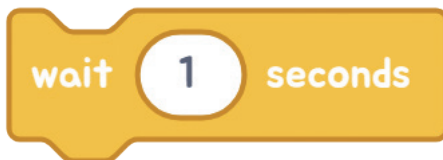


MISSION 2

- 1 Add an LED to make an alarm. When the Siran is on, LED emits the red light unless LED is off.



- 2 Make the LED blink using a wait block



KNOWLEDGE CHECK

1 How can fires impact communities?

- ☐ A Loss of property
- ☐ B Injuries and deaths
- ☐ C Disruption of daily life
- ☐ D All of the above

CORRECT ANSWER ☒ A

2 Why is early fire detection important?

- ☐ A To prevent the spread of fire
- ☐ B To allow for timely evacuation
- ☐ C To reduce property damage
- ☐ D All of the above

CORRECT ANSWER ☒ D

3 How can technology help prevent fires?

- ☐ A By detecting fires early
- ☐ B By improving fire response times
- ☐ C Educating people about fire safety
- ☐ D All of the above

CORRECT ANSWER ☒ D

4 What is a conditional statement used for in programming a fire alarm system?

- ☐ A To determine if a fire has occurred
- ☐ B To trigger an alarm
- ☐ C To control the temperature of the system
- ☐ D To send a notification

CORRECT ANSWER ☒ A

5 How can a fire alarm system help protect the environment?

- ☐ A By preventing fires that can damage forests and wildlife
- ☐ B Reducing the use of harmful chemicals in fire suppression
- ☐ C By promoting energy efficiency in buildings
- ☐ D All of the above

CORRECT ANSWER ☒ A

6 What does a "while loop" do in block coding?

- ☐ A It runs a block of code only once.
- ☐ B It repeats a block of code as long as a condition is true.
- ☐ C It stops the program from running.
- ☐ D It creates a new variable.

CORRECT ANSWER ☒ B

7 What does the "break" block do inside a while loop?

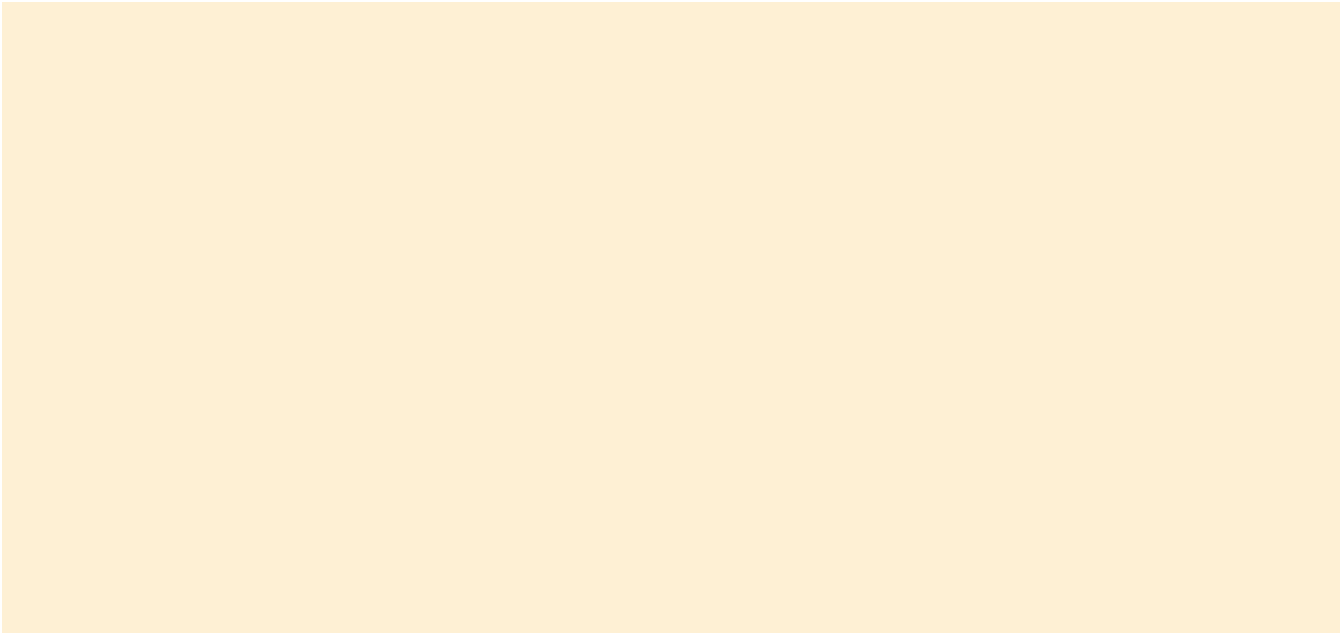
- ☐ A It skips the current iteration of the loop.
- ☐ B It exits the loop immediately.
- ☐ C It creates a new loop.
- ☐ D It changes the loop condition.

CORRECT ANSWER ☒ B



Critical Thinking Challenge

1. Imagine a devastating house fire. Discuss the far-reaching consequences of such a disaster on the community, the environment, and the individuals involved. Consider factors such as property damage, emotional trauma, and long-term impacts.



2. Predict how fire safety technology will evolve in the future. What new innovations or advancements do you foresee? Discuss the potential benefits and challenges of these future developments.

